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Rates Strategy

Deconstructing the EUR yield curve



The functioning of interest rate markets can seem a bit of a black box. What, if anything, is driving interest rate movements at various points of the curve? Where does the value lie for investors and borrowers? How can market participants optimise the risk and returns of their interest rate portfolios? In this report, we lay the foundation of our interest rate relative value framework using the EUR swap curve as an example. After an introduction to our analysis framework, we highlight the main themes and most attractive opportunities.

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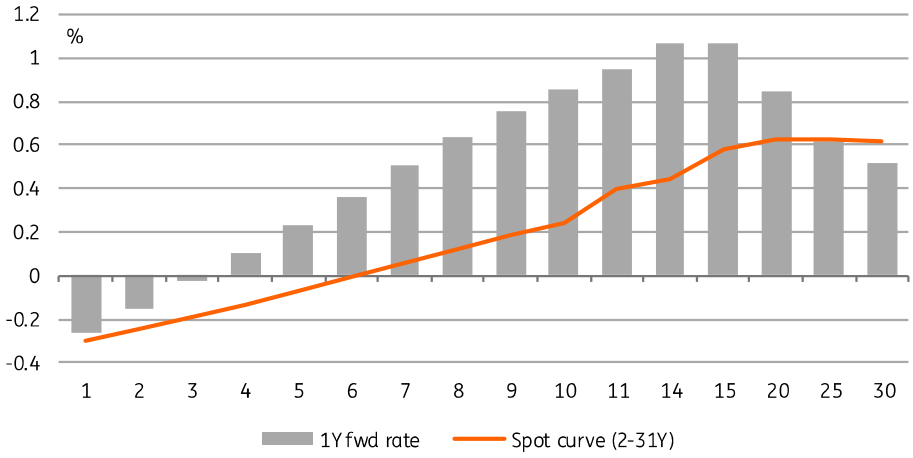
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Introduction: Our RV framework

This publication has a narrow focus: the relative value of interest rates on a single yield curve. In other words, we compare the pricing of rates at various tenors to the curve itself and not relative to other factors or markets (this will be dealt with in a later report). This does not prevent us from trying to explain certain developments with macro themes, but our rich/cheap analysis should be sufficient in itself to highlight trading opportunities.

We base our analysis on a decomposition of the EUR yield curve into non-overlapping forwards. In plain English, the comparison between spot starting rates is flawed in that a significant part of the path for short interest rates implied in spot interest rates is the same. For instance, when comparing the 9Y and 10Y swap rates, we're including the path of short interest rates from 0 to 9 years twice. That information is redundant and can be eliminated using forward rates. To return to our example, it makes more sense to compare the 1Y interest rate in 8Y (8F1Y) to the 1Y interest rate in 9Y (9F1Y) which do not overlap. We show spot and x-years forward 1Y swap rates in the figure below.

EUR swap: spot curve and XF1Y forward rates (%)



Source: Bloomberg, ING

Whilst the forward rates we are looking at are not overlapping, they are strongly correlated. So much so that it is possible to summarise most of the movement of the whole curve from 1F1Y to 30F1Y using only one factor. This factor explains, depending on the timeframe considered, between 97% and 99% of the whole EUR curve's variance. Another way of saying this is that 97% to 99% of the whole curve movement can be explained using just one time series. We term this component the rate level component, rate direction, or principal component 1 (PC1), interchangeably.

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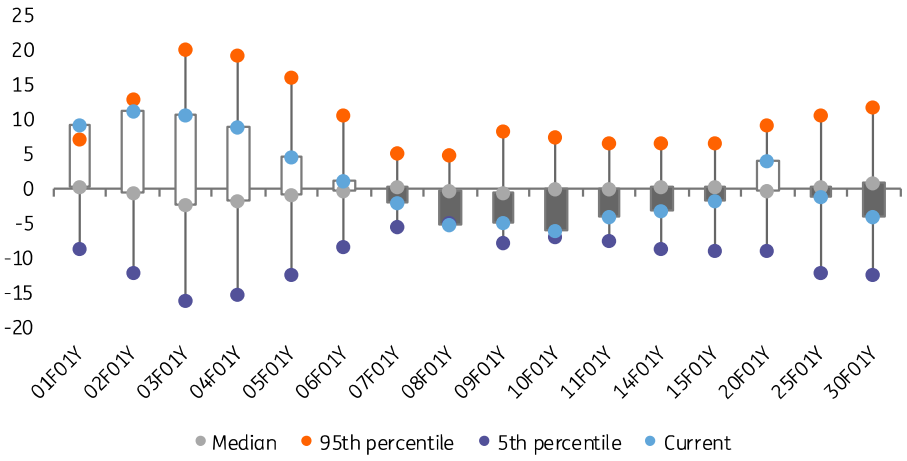
In addition to our approach above, where we sought to remove the path of short interest rate that overlapped between two tenors, we focus on the part of interest rate movements that are not explained by this first component. The main benefit of this approach is to shield us from the noise of general rate movements, and to isolate the developments specific to each tenor. We call PC1 residuals the part of interest movements not explained by rate direction.

Note that the conclusions that we reach about rates in one tenor applies mostly in the context of the whole curve. This approach is thus a good way to identify rates that are too high/too low relative to each other. Hence the name relative value.

Where does value lie on the curve?

Armed with the PC1 residuals of non-overlapping forward rates, we can already draw a number of conclusions about the rate curve. For instance, we can see in Figure 1, showing the current PC1 residual of the EUR swap curve calculated over three years, that short interest rates are currently too high relative to longer ones. As often, it is useful to put these residuals in historical context. Figure 1 shows the current (in blue) level of residuals compared to its 5th (dark blue) and 95th (orange) percentiles, and median value (grey). By this measure, only 1F1Y seems close to its cheapest/highest level in the past three years. Inversely, 10F1Y is close to its richest/lowest level in that period.

Fig 1 Three-year range of PC1 residuals for EUR forward swap rates (bp)

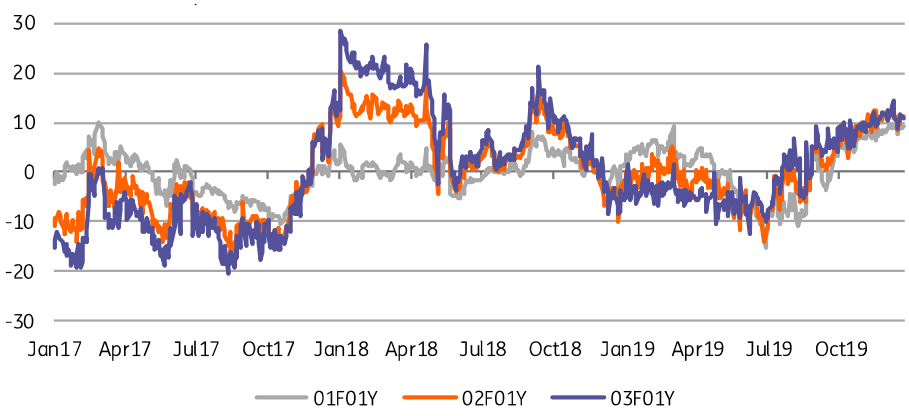


Source: Bloomberg, ING

Cheap/elevated front-end rates are not entirely surprising in the context of improving economic sentiment. EUR rate markets are pricing the possibility of ECB tightening in the coming years, something we discussed a little over a month ago. What elevated PC1 residuals indicate is that they have cheapened relative to other maturities. Implicitly, the curve is not only pricing a more hawkish ECB, it is pricing a change in its reaction function.

One of the advantages of using a long sample period of three years is that it encompasses periods of similar hawkish repricing and allows comparison of the current cheapness to what it was at the time. The historical chart of PC1 residuals in Figure 2 show that short interest rates were at similarly cheap/elevated levels on two occasions: in January and October 2018. Strangely, whilst this was true for 2F1Y and 3F1Y, the 1F1Y rate barely cheapened on both occasions. This makes the recent cheapening of 1F1Y particularly interesting.

Fig 2 Front-end PC1 residuals of selected forward rates (bp)

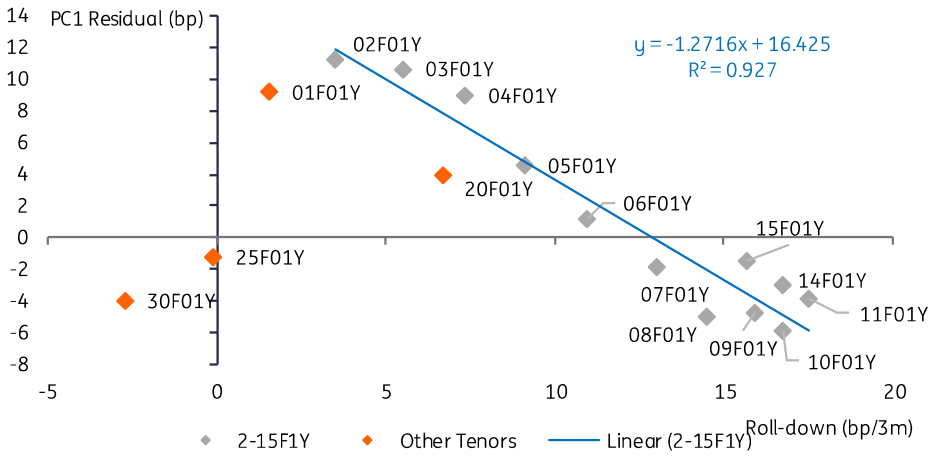


Source: Bloomberg, ING

Value vs Carry: the efficient portfolio frontier

We believe the above is a solid foundation on which to build trade ideas but it is only one step in our thought process. As Figure 2 suggests, the typical timeframe for cheapness/richness to revert is around three months. This makes carry and roll down considerations (carry thereafter) critical in the selection of trade opportunities. Reverting to our example above showing cheap/elevated 1F1Y rate and rich/low 10F1Y, this spread is thus around 14bp too flat. This must be looked at in the context of paying the 1F3Y-10F1Y spread rolling down negatively by 15.5bp every three months. In other words, this trade would breakeven only if the residuals closed in less than 2.6 months. Based on the historical behaviour of this spread, this is not impossible, but investors could be forgiven for looking for a trade where time plays in their favour.

Fig 3 Value vs Carry in EUR forward interest rates



Source: Bloomberg, ING

In our mind, there is a clear trade-off between value (estimated here with PC1 residuals) and carry. In an ideal scenario, investors would select only opportunities where value and carry send the same message. We show in Figure 3 a scatter plot of PC1 residuals against 3m roll-down of swap receiver positions. Based on the reasoning above, investors should have a bias towards paying tenors in the bottom left quadrant (negative residuals and negative roll-down of receivers), and towards receiving tenors in the top right quadrant (positive residual and positive carry of receivers).

This chart also highlights the existence of an 'efficient portfolio frontier' in EUR swap forward rates (blue line in Figure 3) representing the trade-off between carry and valuation. Not all tenors are on that axis, due to difficulty in valuing and trading the

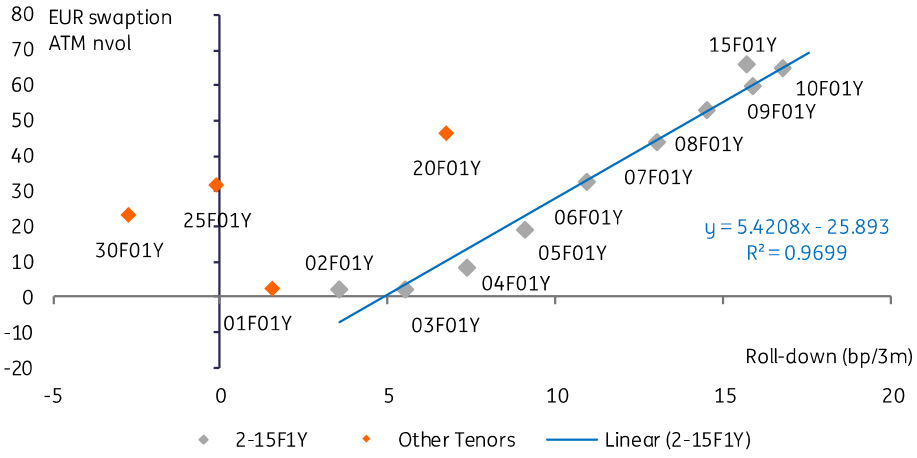
long-dated ones we surmise, but the vast majority are. The negative slope of that linear relationship confirms our intuition that more attractive valuations come at the expense of carry. This also reinforces our earlier observation that current pricing implies residuals reverting to their mean every three months approximately.

As far as trading tenors that are on that 'efficient portfolio frontier' are concerned, this means that the trade-off between value and carry is similar for all tenors on this line. As we allude to above, most investors would be happy to take a view on whether value or carry is more attractive, and thus implicitly on whether this mean reversion assumption is correct.

Volatility reduction and optimisation

One of the insights of our option valuation theory is the trade-off between carry and volatility. In theory, the relationship between the two should be similar to the relationship between valuation and carry that we highlight above. We show this in a scatter plot in Figure 4. Ideally, investors would want to receive (pay) interest rates at tenors where both carry is high (low) and volatility is low (high). Here too, the relationship between roll-down and swaption implied volatility shown in Figure 4, is well defined for most tenors, but breaks down in a few cases, mostly at the long-end. This reinforces our caution in dealing with long-dated forward interest rates for the purpose of relative value analysis.

Fig 4 Carry vs volatility in forward swap rates



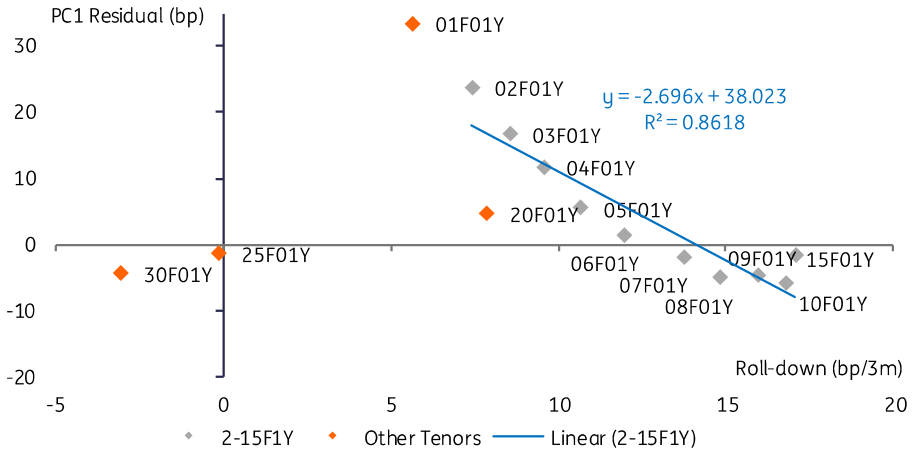
Source: Bloomberg, ING

We state above that most of the movements in non-overlapping forward swap rates can be explained by interest rate direction. Similarly, in implementing curve trades, investors face the risk that most of the performance of their rate portfolios is due to changes in interest rate direction.

We can use our RV analytics to minimise the sensitivity of any curve position to movements in rate direction. We show in Figure 5 the same value vs carry scatter plot of non-overlapping forward swap rates as in Figure 3 but after insulating them from rate direction. We term them direction-neutral forward rates. Note that no rate can be insulated from rate direction on its own. This chart reflects the weighting that would ensure that any spread trade is direction-neutral.

The general shape of the value vs carry relationship remains the same, but the slope of the efficient portfolio frontier has doubled. This suggests that the trade-off between value and carry described above is almost twice as attractive as before minimising sensitivity to rate direction. This makes intuitive sense, as the lower volatility requires less carry compensation.

Fig 5 Carry vs volatility of direction-neutral non-overlapping forward interest rates

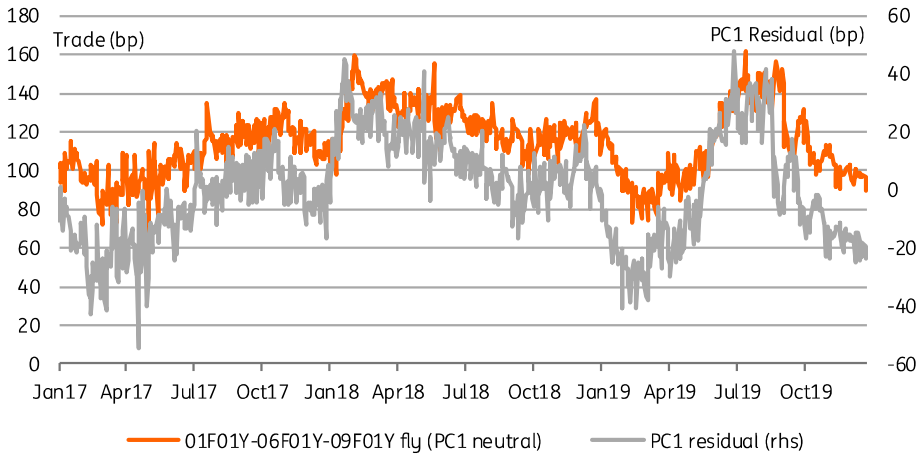


Source: Bloomberg, ING

A look at Figure 5 shows that for most of tenors except long-dated ones, spread trades entail taking a view on value vs carry. Even if in direction-neutral terms, value is more attractive than carry than in direction-exposed terms, we suspect many investors would find steepeners, such as paying 1F1Y-10F1Y, unpalatable due to their negative carry.

Butterflies, on the other hand, offer a finer way to position on the curve as they can be understood as a combination of a steepener and a flattener. Based on Figure 5, we identify the 1F1Y-6F1Y-9F1Y butterfly as an attractive paying opportunity with 1F1Y and 9F1Y above or close to the efficient portfolio frontier, while 6Y1Y is below it. Based on our analysis, the belly is 23bp too expensive in direction-neutral terms. This compares favourably to a negative carry of -2bp/3m, in our view. This implies this trade would need to revert to its mean in more than 36 months for carry to eat out the profit.

Fig 6 1F1Y-6F1Y-9F1Y direction-neutral butterfly: rate and PC1 residuals



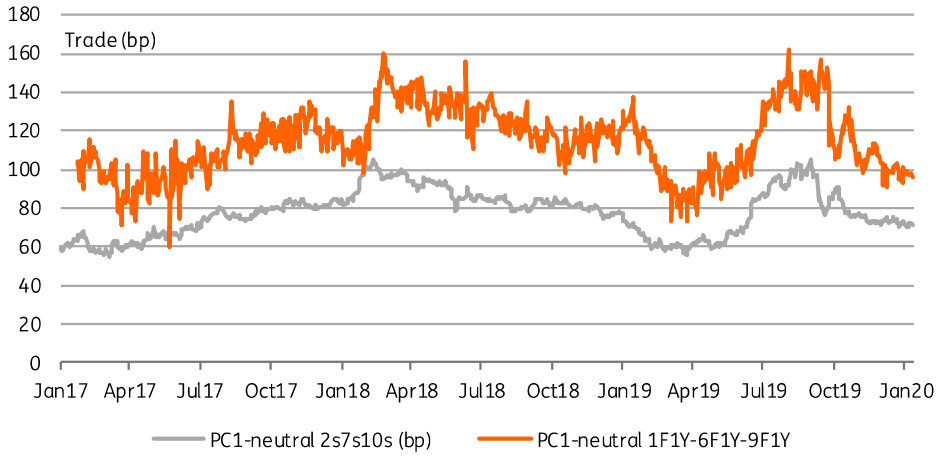
Source: Bloomberg, ING

As shown in Figure 6, this structure has had a tendency to cheapen when interest rate markets priced a more hawkish ECB (eg, in 1Q18), or when the long-end flattened aggressively (in 2Q19). That the 1F1Y-6F1Y-9F1Y direction-neutral butterfly has not cheapened with rate markets pricing a less dovish ECB reaction function going forward is something we disagree with. As we saw in Figure 2, previous hawkish ECB re-pricing did not necessarily entail a re-pricing of 1F1Y. Arguably, the economy is in a worse place now than it was at the start of 2018 so the odds of a near-term hike should be fairly low. It is more correct, in our view, for the belly of the curve (eg, the 6F1Y) to reflect a better macro backdrop.

From forward to spot interest rates

Our framework described above allows us to carry out a finer analysis of the EUR swap curve than relying only on spot rates. We have identified the 1F1Y-6F1Y-9F1Y direction-neutral butterfly as offering an attractive trade-off between valuation and carry. Its equivalent in spot interest rate is the direction-neutral 2s7s10s butterfly (shown in Figure 7 compared to 1F1Y-6F1Y-9F1Y). In addition to having a very strong correlation with the forward butterfly we highlighted above, this version has a positive carry and roll of +0.7bp/m, and is 9bp away from its three year mean.

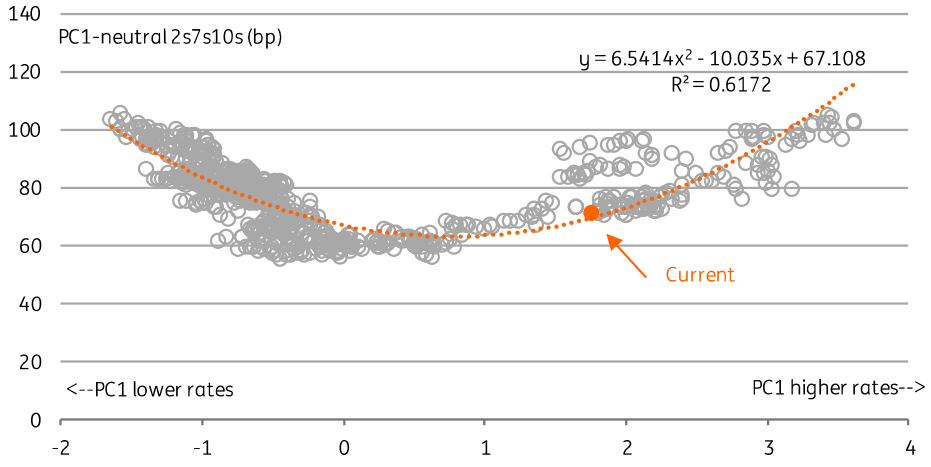
Fig 7 Direction-neutral butterfly 2s7s10s butterfly vs 1F1Y-6F1Y-9F1Y



Source: Bloomberg, ING

The spot version of this trade is, at first glance, an even better deal than the forward version: it shares many of its characteristics with the added benefit of a slightly positive carry. The downside of the spot version is not obvious at first. When plotted against rate direction (PC1) as in Figure 8, it appears that the direction-neutral 2s7s10s butterfly has in reality a non-linear relationship with rates: before a certain inflexion point, it cheapens when rates fall, and after that point it cheapens when rates rise.

Fig 8 Direction-neutral 2s7s10s butterfly against rates direction



Source: Bloomberg, ING

We feel less confident in saying that this non-linear relationship will continue in the future although the inflection point appears to coincide with spot 7Y reaching 0%. In any case, since 7Y EUR swaps are currently flirting with that level, this would be an additional reason to pay this butterfly.

Trade summary

- **The trade:** pay direction neutral EUR 2s7s10s fly (3-2-0.84 weights)
- **Entry point:** 71bp (including 0.5bp charge from 70.5bp mid at the time of writing)
- **Target:** 91bp
- **Stop loss:** 61bp
- **Carry and roll:** -2.5bp/3m and +3.2bp/3m respectively, total +0.7bp/3m
- **Horizon:** end-2020
- **Main risk(s):** Actual change in the ECB reaction function that would see it raise interest rates this year or next

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